

1. 利用柯西收敛原理证明:

(1) 级数 $\sum_{n=1}^{\infty} \frac{\cos n}{n(n+1)}$ 收敛; (2) 级数 $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ 发散;

(3) 设两个级数 $\sum_{n=1}^{\infty} a_n$ 与 $\sum_{n=1}^{\infty} b_n$ 都收敛, 且存在正整数 N , 使当 $n \geq N$ 时有 $a_n \leq u_n \leq b_n$, 则级数 $\sum_{n=1}^{\infty} u_n$ 也收敛.

1. (1). $a_n = \frac{\cos n}{n(n+1)} = \cos n \left(\frac{1}{n} - \frac{1}{n+1} \right)$

$$|S_{n+p} - S_n| = \left| \sum_{n+1}^{n+p} \cos n \left(\frac{1}{n} - \frac{1}{n+1} \right) \right| \leq \left| \sum_{n+1}^{n+p} \left(\frac{1}{n} - \frac{1}{n+1} \right) \right| = \left| \frac{1}{n+1} - \frac{1}{n+p+1} \right| \leq \frac{1}{n+1} < \varepsilon$$

令 $\varepsilon = \left[\frac{1}{n} \right] - 1$ 即可

(2) $|S_{n+p} - S_n| = \sum_{n+1}^{n+p} \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n+p}} \geq \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+p}$

令 $p = n+1$, 有 $\sum_{n+1}^{n+p} \frac{1}{\sqrt{n}} \geq \frac{1}{n+1} + \dots + \frac{1}{2n+1} \leq (n+1) \frac{1}{n+1} = 1$

故级数发散

(3) $\sum_{n=1}^{\infty} a_n$ 收敛, 则 $\exists N_1$, 使得当 $n > N_1$ 时, $\left| \sum_{k=n}^{n+p} a_k \right| < \varepsilon$

$\exists N_2$, $n > N_2$ 时, $\left| \sum_{k=n}^{n+p} b_k \right| < \varepsilon$

取 $N_0 = \max\{N, N_1, N_2\}$

$n > N_0$ 时, $-\varepsilon \leq \sum_{k=n}^{n+p} a_k \leq \sum_{k=n}^{n+p} u_k \leq \sum_{k=n}^{n+p} b_k \leq \varepsilon$

则 $\left| \sum_{k=n}^{n+p} u_k \right| \leq \varepsilon$

证

2. 已知级数 $\sum_{n=1}^{\infty} a_n$ 收敛, 又知级数 $\sum_{n=1}^{\infty} b_n$ 发散, 问级数 $\sum_{n=1}^{\infty} (a_n \pm b_n)$ 是否收敛?

3. 判断下列级数是否收敛:

(1) $\sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})$; (2) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$;

(3) $\sum_{n=1}^{\infty} \frac{1}{(5n-4)(5n+1)}$; (4) $\sum_{n=1}^{\infty} \cos^2 \frac{\pi}{n}$;

(5) $\sum_{n=1}^{\infty} \frac{n}{2n-1}$; (6) $\sum_{n=1}^{\infty} \frac{\ln^n 5}{3^n}$;

(7) $\sum_{n=1}^{\infty} \sqrt[n]{0.0001}$; (8) $\sum_{n=1}^{\infty} \frac{1}{6n}$.

2. 不收敛

由于 $\sum_{n=1}^{\infty} a_n$ 收敛, 则 $\lim_{n \rightarrow \infty} a_n = 0$

而 $\sum_{n=1}^{\infty} b_n$ 发散, 则 $\lim_{n \rightarrow \infty} b_n \neq 0$

则 $\lim_{n \rightarrow \infty} (a_n \pm b_n) \neq 0$,

故级数一定发散

3. (1) $S_n = \sqrt{n+1} - 1$ $\lim_{n \rightarrow \infty} S_n = \infty$, 故发散.

(2) $\sum_{n=1}^{\infty} (\frac{1}{2n-1} - \frac{1}{2n+1}) \frac{1}{2}$, $S_n = \frac{1}{2} \cdot (1 - \frac{1}{2n+1})$, $\lim_{n \rightarrow \infty} S_n = \frac{1}{2}$, 收敛.

(3) $\sum_{n=1}^{\infty} (\frac{1}{5n-4} - \frac{1}{5n+1}) \frac{1}{5}$, $S_n = \frac{1}{5} \cdot (1 - \frac{1}{5n+1})$, $\lim_{n \rightarrow \infty} S_n = \frac{1}{5}$ 收敛

(4) $\lim_{n \rightarrow \infty} \cos^2 \frac{2\pi}{n} = \cos^2 0 = 1$, 发散 $\frac{1 + \cos \frac{2\pi}{n}}{2}$

(5) $\lim_{n \rightarrow \infty} \frac{n}{2n-1} = \frac{1}{2}$, 发散.

(6) $\sum_{n=1}^{\infty} (\frac{\ln 5}{3})^n$, 而 $\frac{\ln 5}{3} < 1$, 故收敛

(7) $\sum_{n=1}^{\infty} (\frac{1}{10000})^{\frac{1}{n}}$ $\lim_{n \rightarrow \infty} (\frac{1}{10000})^{\frac{1}{n}} = 1$, 发散

(8) $\sum_{n=1}^{\infty} \frac{1}{6n}$ $\sum_{k=n}^{n+p} \frac{1}{6k} = \frac{1}{6} \cdot (\frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{n+p})$, 取 $p=n$, 有
 原式 = $\frac{1}{6} \cdot n \cdot \frac{1}{2n} = \frac{1}{12}$.

故发散

4. 设级数 $\sum_{n=1}^{\infty} u_n$ 的部分和序列为 $\{S_n\}$. 若 $n \rightarrow \infty$ 时 $\{S_{2n}\}$ 与 $\{S_{2n+1}\}$ 都收敛且收敛

到同一个常数 A , 证明级数 $\sum_{n=1}^{\infty} u_n$ 收敛.

5. 设级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 且 $u_n \geq u_{n+1} \geq 0 (n = 1, 2, \dots)$, 证明:

$$\lim_{n \rightarrow \infty} n u_n = 0.$$

(提示: 利用本节定理 4, 先证 $n u_{2n} \rightarrow 0$, 因而 $2n u_{2n} \rightarrow 0$. 再证 $(2n+1) u_{2n+1} \rightarrow 0$.)

4. 当 $n > N_1$ 时, $|S_{2n} - A| < \frac{\varepsilon}{2}$ $n > N_1 \Rightarrow 2n > 2N_1$, 令 $2n = n'$.

当 $n > N_2$ 时, $|S_{n+1} - A| < \frac{\varepsilon}{2}$ $n > N_2 \Rightarrow 2n+1 > 2N_2+1$, 令 $2n+1 = n'$.

取 $N = \max\{2N_1, 2N_2+1\}$, 则 $n > N$ 时, $|S_n - A| < \frac{\varepsilon}{2}$.

$$\therefore \left| \sum_{k=n+1}^{n+p} u_k \right| = |S_{n+p} - S_n| = |S_{n+p} - A - (S_n - A)| \leq |S_{n+p} - A| + |S_n - A| = \varepsilon$$

故 $\sum_{k=n+1}^{n+p} u_k$ 可以充分小,

故 $\sum_{n=1}^{\infty} u_n$ 收敛

5. $\because \sum_{n=1}^{\infty} u_n$ 收敛, 则 $\forall \varepsilon > 0, \exists N > 0$, $\left| \sum_{k=n+1}^{n+p} u_k \right| < \varepsilon$

即任取 $p > 1$, 都有 $\lim_{n \rightarrow \infty} \sum_{k=n+1}^{n+p} u_k = 0$

令 $p = n$, 即有 $\lim_{n \rightarrow \infty} (u_{n+1} + u_{n+2} + \dots + u_{2n}) = 0$

而 $u_{n+1} + u_{n+2} + \dots + u_{2n} \geq n \cdot u_{2n} \geq 0$.

由夹逼定理, $\lim_{n \rightarrow \infty} n \cdot u_{2n} = 0$, 则 $\lim_{n \rightarrow \infty} 2n \cdot u_{2n} = 0$

而 $(2n+1) u_{2n+1} = 2n \cdot u_{2n+1} + u_{2n+1} \leq 2n \cdot u_{2n} + u_{2n+1}$,

$$0 \leq \lim_{n \rightarrow \infty} (2n+1) u_{2n+1} \leq \lim_{n \rightarrow \infty} (2n \cdot u_{2n} + u_{2n+1}) = 0$$

故 $\lim_{n \rightarrow \infty} (2n+1) u_{2n+1} = 0$

故 $\lim_{n \rightarrow \infty} n \cdot u_n = 0$

1. 讨论下列级数的敛散性:

(1) $\sum_{n=1}^{\infty} 2^n \sin \frac{\pi}{4^n};$

(2) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n^3+1}};$

(3) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n}};$

(4) $\sum_{n=1}^{\infty} \frac{4n}{n^2+4n-3};$

(5) $\sum_{n=1}^{\infty} \frac{n^n}{(n^2+3n+1)^{\frac{n+2}{2}}};$

(6) $\sum_{n=2}^{\infty} \frac{n}{(\ln n)^{\ln n}}$ (提示: n 充分大后, $(\ln n)^{\ln n} > n^3$);

(7) $\sum_{n=1}^{\infty} n \tan \frac{1}{2^n}.$

1. (1) 发散. 令 $a_n = 2^n \sin \frac{\pi}{4^n}, b_n = 2^n$

则 $\frac{a_n}{b_n} = \sin \frac{\pi}{4^n} \in (0, 1)$

又有 b_n 发散, 则 a_n 发散

(2) 收敛

$0 < a_n = \frac{1}{\sqrt{2n^3+1}} \leq \frac{1}{\sqrt{2n^3}} = \frac{1}{\sqrt{2} \cdot n^{\frac{3}{2}}} = b_n$

而 $\sum_{n=1}^{\infty} b_n$ 收敛

故 $\sum_{n=1}^{\infty} a_n$ 收敛

(3) 发散 $\frac{1}{\sqrt[n]{n}} \geq \frac{1}{n}$. 而 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 故 $\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n}}$ 发散

(4) 发散 $a_n = \frac{4n}{n^2+4n-3}$ 取 $b_n = \frac{1}{n}$, $\frac{a_n}{b_n} = \frac{4n^2}{n^2+4n-3}$. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 4$
而 b_n 发散, 故 a_n 发散.

(5) 收敛 $\frac{n^n}{(n^2+3n+1)^{\frac{n+2}{2}}} \leq \frac{n^n}{n^{n+2}} = \frac{1}{n^2}$ 而 $\frac{1}{n^2}$ 收敛

故收敛

(6) 收敛 $(\ln n)^{\ln n} = e^{\ln n \cdot \ln \ln n} = n^{(\ln \ln n)}$ $\lim_{n \rightarrow \infty} \frac{n^{\ln(\ln n)}}{n^3} = \lim_{n \rightarrow \infty} n^{\ln(\ln n) - 3} = \infty$

当 n 充分大时, $(\ln n)^{\ln n} > n^3$

故 $\exists M > 0$, 当 $n > N$ 时, $\frac{n}{(\ln n)^{\ln n}} < \frac{n}{n^3} = \frac{1}{n^2}$, 而 $\sum_{n=2}^{\infty} \frac{1}{n^2}$ 收敛

故收敛

(7) 收敛 $\sum_{n=1}^{\infty} n \cdot \tan \frac{1}{3^n}$ 令 $b_n = \frac{n}{3^n}$, $\lim_{n \rightarrow \infty} \frac{n \cdot \tan \frac{1}{3^n}}{\frac{n}{3^n}} = 1$

而 $\sqrt[n]{\frac{n}{3^n}} = \frac{\sqrt[n]{n}}{3}$, $\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n}}{3} = \frac{1}{3}$, 故 $\sum_{n=1}^{\infty} \frac{n}{3^n}$ 收敛

故 $\sum_{n=1}^{\infty} n \tan \frac{1}{3^n}$ 收敛

2. 讨论下列级数的敛散性:

$$(1) \sum_{n=1}^{\infty} \frac{n^5}{n!};$$

$$(2) \sum_{n=1}^{\infty} \frac{n!}{3n^2};$$

$$(3) \sum_{n=1}^{\infty} \frac{3^n \cdot n!}{n^n};$$

$$(4) \sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}};$$

$$(5) \sum_{n=1}^{\infty} \frac{n^2}{\left(3 - \frac{1}{n}\right)^n};$$

$$(6) \sum_{n=2}^{\infty} \frac{n}{(\ln n)^n};$$

$$(7) \sum_{n=1}^{\infty} \frac{1000^n}{n!};$$

$$(8) \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!};$$

$$(1) a_n = \frac{n^5}{n!}, \quad \frac{a_n}{a_{n-1}} = \frac{n^5}{n!} \frac{(n-1)!}{(n-1)^5} = \frac{n^4}{(n-1)^5}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = 0$$

故 $\sum_{n=1}^{\infty} \frac{n^5}{n!}$ 收敛

$$(2) a_n = \frac{n!}{3n^2}, \quad \frac{a_n}{a_{n-1}} = \frac{n!}{3n^2} \frac{3(n-1)^2}{(n-1)!} = \frac{n(n-1)^2}{n^2} \frac{(n-1)^2}{n}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = \infty$$

故 $\sum_{n=1}^{\infty} \frac{n!}{3n^2}$ 发散

$$(3) a_n = \frac{3^n \cdot n!}{n^n}, \quad \frac{a_n}{a_{n-1}} = 3n \cdot \frac{(n-1)^{n-1}}{n^n} = 3 \cdot \left(\frac{n-1}{n}\right)^{n-1}, \quad \lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right)^{n-1} = \lim_{n \rightarrow \infty} e^{(n-1) \ln \frac{n-1}{n}} = \lim_{n \rightarrow \infty} e^{\frac{1-n}{n}} = e^{-1}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = \frac{3}{e} > 1,$$

故发散

$$(4) a_n = \frac{1}{n\sqrt{n}}, \quad b_n = \frac{1}{n}, \quad \frac{a_n}{b_n} = \frac{1}{\sqrt{n}}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$$

故 $\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}$ 收敛

$$(5) a_n = \frac{n^2}{\left(3 - \frac{1}{n}\right)^n}, \quad \sqrt[n]{a_n} = \frac{n^{\frac{2}{n}}}{3 - \frac{1}{n}}, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \frac{1}{3}, \quad \text{故}$$

原级数收敛

$$(6) \sum_{n=2}^{\infty} \frac{n}{(\ln n)^n}, \quad \sqrt[n]{a_n} = \frac{\sqrt[n]{n}}{\ln n}, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 0$$

故级数收敛

$$(7) a_n = \frac{1000^n}{n!}, \quad \frac{a_n}{a_{n-1}} = \frac{1000}{n}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = 0$$

故级数收敛

$$(8) a_n = \frac{(n!)^2}{(2n)!}, \quad \frac{a_n}{a_{n-1}} = \frac{n^2}{2n(2n-1)} = \frac{n}{4n-2}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = \frac{1}{4}$$

故级数收敛

$$(9) \sum_{n=1}^{\infty} \frac{1}{3^n} \left(\frac{n+1}{n} \right)^{n^2};$$

$$(10) \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p} \quad (p > 0) \text{ (提示: 用积分判别法);}$$

$$(11) \sum_{n=3}^{\infty} \frac{1}{n(\ln \ln n)^q} \quad (q > 0) \text{ (提示: } n \text{ 充分大后, } (\ln \ln n)^q < (\ln n)^{1/2} \text{);}$$

$$(12) \sum_{n=3}^{\infty} \frac{1}{n(\ln n)^p (\ln \ln n)^q} \quad (p > 0, q > 0) \text{ (提示: 利用积分判别法并分别讨论下}$$

面 4 种情况: ① $p > 1$; ② $p < 1$; ③ $p = 1, q > 1$; ④ $p = 1, q \leq 1$).

$$(9) \quad a_n = \frac{1}{3^n} \cdot \left(\frac{n+1}{n} \right)^{n^2} \quad n\sqrt{a_n} = \frac{1}{3} \cdot \left(1 + \frac{1}{n} \right)^n \quad \lim_{n \rightarrow \infty} n\sqrt{a_n} = \frac{e}{3} < 1$$

故级数收敛

$$(10) \quad f(n) = \frac{1}{n \cdot (\ln n)^p} \quad (n \geq 2), \quad f(n) \text{ 单调}$$

$$\int_2^{\infty} f(n) dn = \lim_{A \rightarrow \infty} \int_2^A \frac{1}{n(\ln n)^p} dn = \lim_{A \rightarrow \infty} \int_2^A \frac{1}{(\ln n)^p} d(\ln n) = \lim_{A \rightarrow \infty} \left[\frac{1}{1-p} (\ln n)^{1-p} \right] \Big|_2^A = \lim_{A \rightarrow \infty} \frac{1}{1-p} [(\ln A)^{1-p} - (\ln 2)^{1-p}]$$

① 若 $0 < p < 1$, $\int_2^{\infty} f(n) dn$ 不收敛, 级数发散

② 若 $p \geq 1$, $\int_2^{\infty} f(n) dn = \frac{(\ln 2)^{1-p}}{p-1}$, 级数收敛

$$(11) \quad \text{若 } (\ln \ln n)^q < (\ln n)^{\frac{1}{2}}, \quad \text{令 } x = \ln n, \quad n \text{ 与 } x \text{ 一一对应, } n \rightarrow \infty, x \rightarrow \infty$$

$$\text{只要 } (\ln x)^q < \sqrt{x}, \quad \frac{(\ln x)^q}{\sqrt{x}} < 1,$$

$$\text{而 } \lim_{x \rightarrow \infty} \frac{(\ln x)^q}{\sqrt{x}} = 0, \quad \text{即 } \exists N > 0, \text{ 当 } n > N \text{ 时, } \forall \varepsilon > 0,$$

$$\frac{(\ln x)^q}{\sqrt{x}} < \varepsilon,$$

即当 n 足够大时, $(\ln \ln n)^q < (\ln n)^{\frac{1}{2}}$ 恒成立

$$\frac{1}{n(\ln \ln n)^q} > \frac{1}{n(\ln n)^{\frac{1}{2}}}, \quad \text{由 (10), } \sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}} \text{ 发散}$$

故原级数发散

(12)

(12) $\sum_{n=3}^{\infty} \frac{1}{n(\ln n)^p (\ln \ln n)^q}$ ($p > 0, q > 0$) (提示: 利用积分判别法并分别讨论下面 4 种情况: ① $p > 1$; ② $p < 1$; ③ $p = 1, q > 1$; ④ $p = 1, q \leq 1$).

$$\because n > 3, \therefore \ln n > 1$$

① $p > 1$ 时, $\frac{1}{n(\ln n)^p (\ln \ln n)^q} < \frac{1}{n(\ln n)^p}$, 原级数收敛

② $0 < p < 1$ 时 $\frac{1}{n(\ln n)^p (\ln \ln n)^q} > \frac{1}{n(\ln \ln n)^q}$, 原级数发散

③ $p = 1$ 时

$$f(n) = \frac{1}{n \cdot \ln n \cdot (\ln \ln n)^q}, \quad \int f(n) dn = \int \frac{1}{\ln n (\ln \ln n)^q} d \ln n \xrightarrow{t = \ln n} \int \frac{1}{t (\ln t)^q} dt = \int \frac{1}{(\ln t)^q} d \ln t = \frac{1}{1-q} (\ln t)^{1-q}$$

$$\int_3^{\infty} f(n) dn = \frac{1}{1-q} (\ln \ln n)^{1-q} \Big|_3^{\infty} = \lim_{A \rightarrow \infty} \left[\frac{1}{1-q} (\ln \ln A)^{1-q} - \frac{1}{1-q} (\ln \ln 3)^{1-q} \right]$$

(i) $q > 1$ 时, $\int_3^{\infty} f(n) dn$ 收敛, 级数收敛

(ii) $0 < q \leq 1$ 时, $\int_3^{\infty} f(n) dn$ 发散, 级数发散

3. 证明: 若正项级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 则级数 $\sum_{n=1}^{\infty} u_n^2$ 也收敛. 反之不一定成立, 试举例说明

$$\because \sum_{n=1}^{\infty} u_n \text{ 收敛}, \quad \lim_{n \rightarrow \infty} u_n = 0, \quad \text{则 } \exists N > 0, \text{ 当 } n > N \text{ 时}, 0 < u_n < 1$$

$$\text{则必有 } n > N \text{ 时}, u_n^2 < u_n$$

$$\text{故 } \sum_{n=1}^{\infty} u_n^2 \text{ 收敛}$$

$$\text{反例: } \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ 收敛}$$

$$\text{而 } \sum_{n=1}^{\infty} \frac{1}{n} \text{ 发散}$$

4. 证明: 若级数 $\sum_{n=1}^{\infty} a_n^2$ 与 $\sum_{n=1}^{\infty} b_n^2$ 都收敛, 则级数

$$\sum_{n=1}^{\infty} |a_n b_n|, \quad \sum_{n=1}^{\infty} (a_n + b_n)^2, \quad \sum_{n=1}^{\infty} \frac{|a_n|}{n}$$

也收敛.

5. 若正项级数 $\sum_{n=1}^{\infty} u_n$ 与 $\sum_{n=1}^{\infty} v_n$ 都发散, 问下列级数是否发散?

$$(1) \sum_{n=1}^{\infty} (u_n + v_n); \quad (2) \sum_{n=1}^{\infty} (u_n - v_n); \quad (3) \sum_{n=1}^{\infty} u_n v_n.$$

6. 设 $\lim_{n \rightarrow \infty} n u_n = l$, 其中 $0 < l < +\infty$. 证明: 级数 $\sum_{n=1}^{\infty} u_n^2$ 收敛, 而 $\sum_{n=1}^{\infty} u_n$ 发散.

$$4. \textcircled{1} |a_n b_n| \leq \frac{1}{2} (a_n^2 + b_n^2) \quad \textcircled{2} (a_n + b_n)^2 \leq 2(a_n^2 + b_n^2)$$

$$\text{而级数 } \sum_{n=1}^{\infty} a_n^2, \sum_{n=1}^{\infty} b_n^2 \text{ 收敛} \quad \text{而 } \sum_{n=1}^{\infty} 2(a_n^2 + b_n^2) \text{ 收敛}$$

$$\text{故 } \sum_{n=1}^{\infty} |a_n b_n| \text{ 收敛} \quad \text{故 } \sum_{n=1}^{\infty} (a_n + b_n)^2 \text{ 收敛}$$

$$\textcircled{3} |a_n| \cdot \frac{1}{n} \leq \sqrt{a_n^2} \cdot \sqrt{\frac{1}{n^2}}, \text{ 收敛}$$

5. (1) 发散

(2) 不确定

(3) 不确定.

6. $\lim_{n \rightarrow \infty} \frac{u_n}{\frac{1}{n}} = l \in (0, +\infty)$, 而级数 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 则 $\sum_{n=1}^{\infty} u_n$ 发散

$\because \lim_{n \rightarrow \infty} n u_n = l$, 则 $\exists M > 0$, 当 $n > N$ 时, $\forall \varepsilon > 0$, $|n u_n - l| < \varepsilon$

$$|n u_n - l| > |n u_n| - |l|, \quad \therefore |n u_n| < \varepsilon + |l|$$

$$\text{取 } \varepsilon = \frac{|l|}{2}, \quad |n u_n| < \frac{3|l|}{2}.$$

$$n^2 u_n^2 < \frac{9l^2}{4}.$$

$$\therefore u_n^2 < \frac{9l^2}{4n^2}, \quad \text{级数 } \sum_{n=1}^{\infty} \frac{9l^2}{4n^2} \text{ 收敛}$$

$$\text{故级数 } \sum_{n=1}^{\infty} u_n^2 \text{ 收敛}$$